

# Background

- **Autism Spectrum Disorder (ASD)** is a common developmental disorder that can affect the way people learn, interact, and behave (National Institute of Mental Health, 2024)
- There is no cure
- **Diet** has been seen to have an effect on the symptoms exhibited (MacFabe, 2012)
- ***Drosophila melanogaster*** is a commonly used organism which models the human brain due to their shared genes with humans (Tian et al, 2017)



Human brain model: *Drosophila melanogaster*

- *Drosophila* are social organisms and prefer to stay in groups
- **Citric and benzoic acid** are the most commonly used **food preservatives**, but little tests have been run with them

## Research Question

Does the intake of the common food preservative acids (FPAs) citric acid and benzoic acid have a positive effect on the symptoms of Autism Spectrum Disorder (ASD) exhibited by a *Drosophila melanogaster* model?

## Hypothesis

It is hypothesized that when *Drosophila* with ASD consume citric acid, the citric acid will worsen social interaction due to studies in which citric acid has been shown to display inflammatory properties. Additionally it is hypothesized that a benzoic acid diet will improve the social interaction of *Drosophila* due to a previous study in which benzoic acid improved gut function in piglets.

## Variables

IV

Type of food being fed to the *Drosophila* (Regular food, Lead food or Citric/Benzoic Acid food)

DV

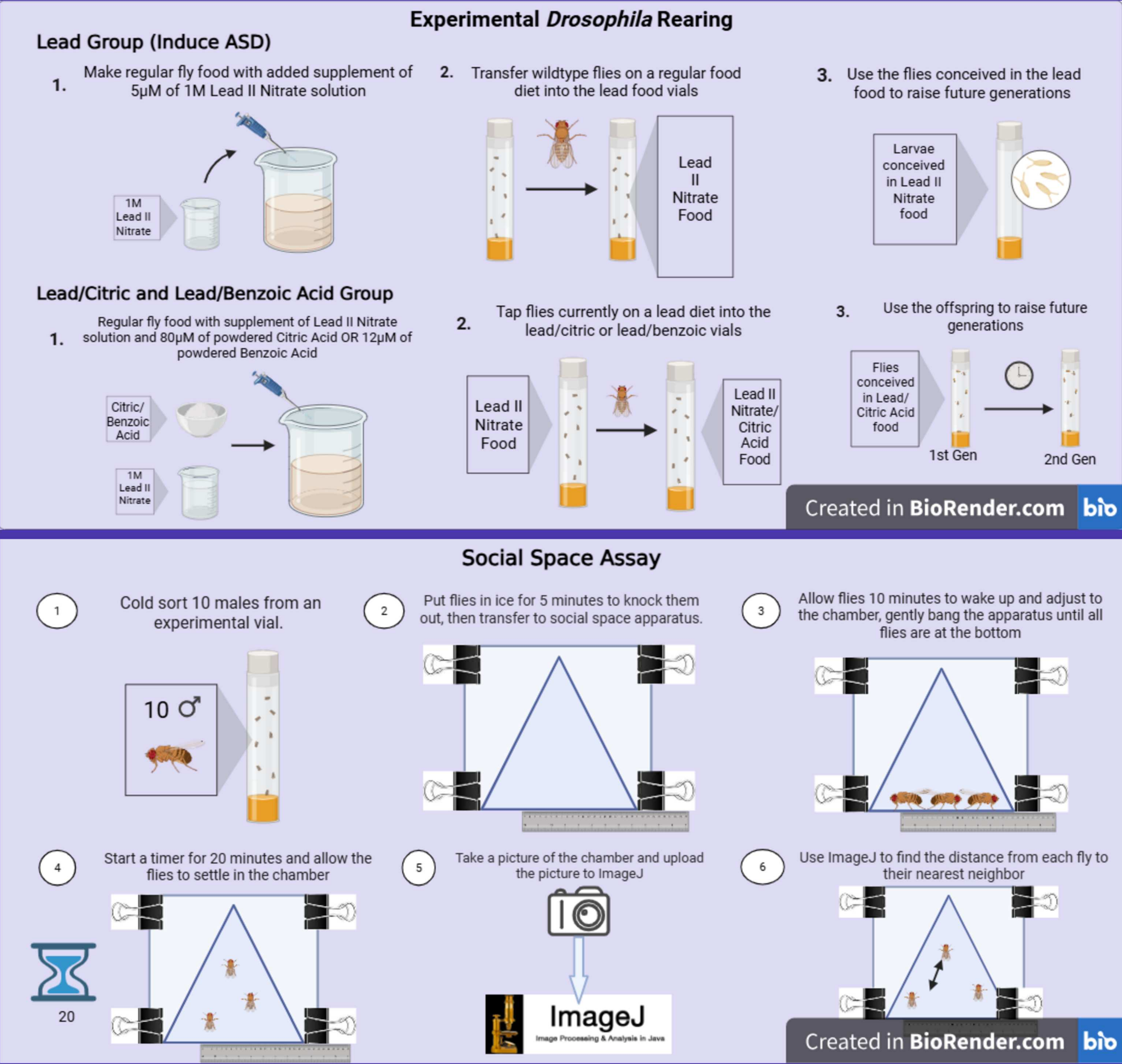
Severity of the lack of social interaction exhibited (Distance between each fly and their nearest neighbor)

Control

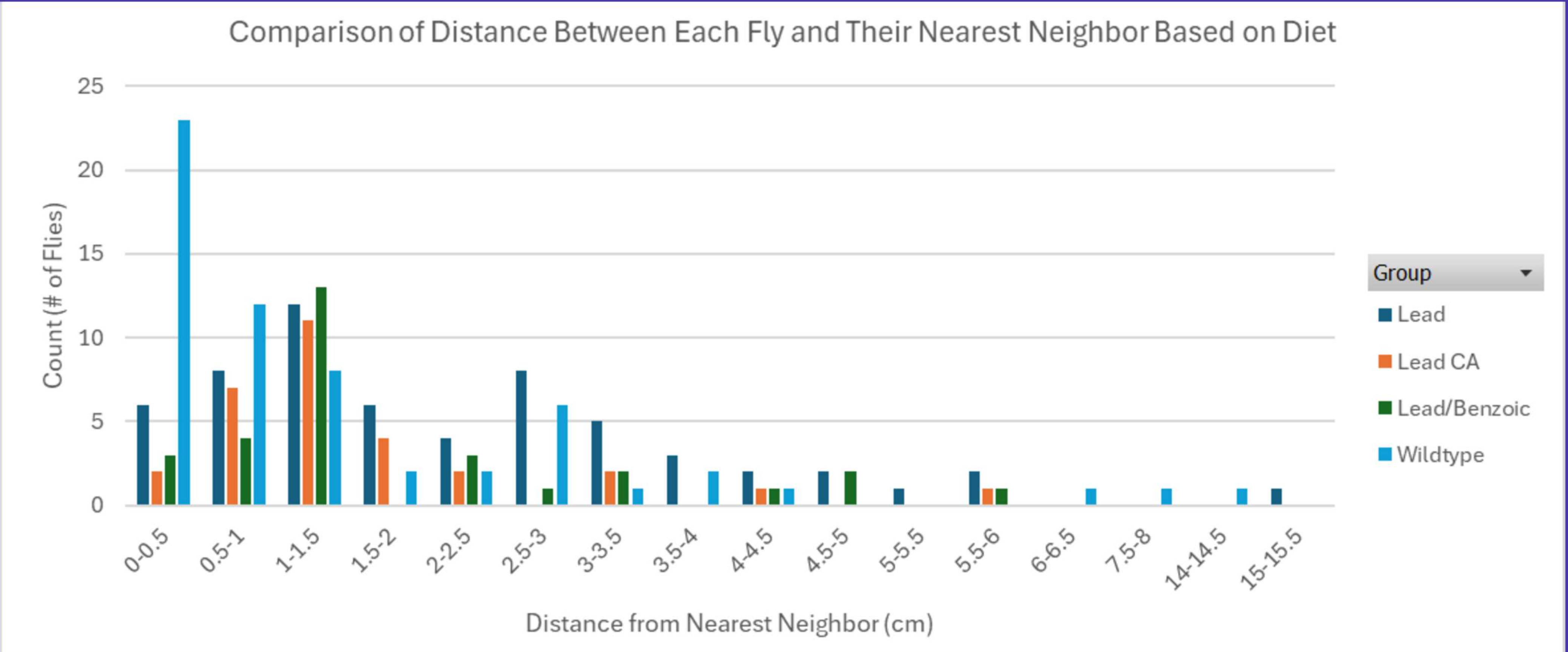
Wildtype and ASD-induced *Drosophila* that are fed regular fly food

# The Effect of Citric Acid and Benzoic Acid on the Severity of ASD-Like Symptoms Exhibited by a Lead-Induced ASD Model of *Drosophila melanogaster*

## Methods

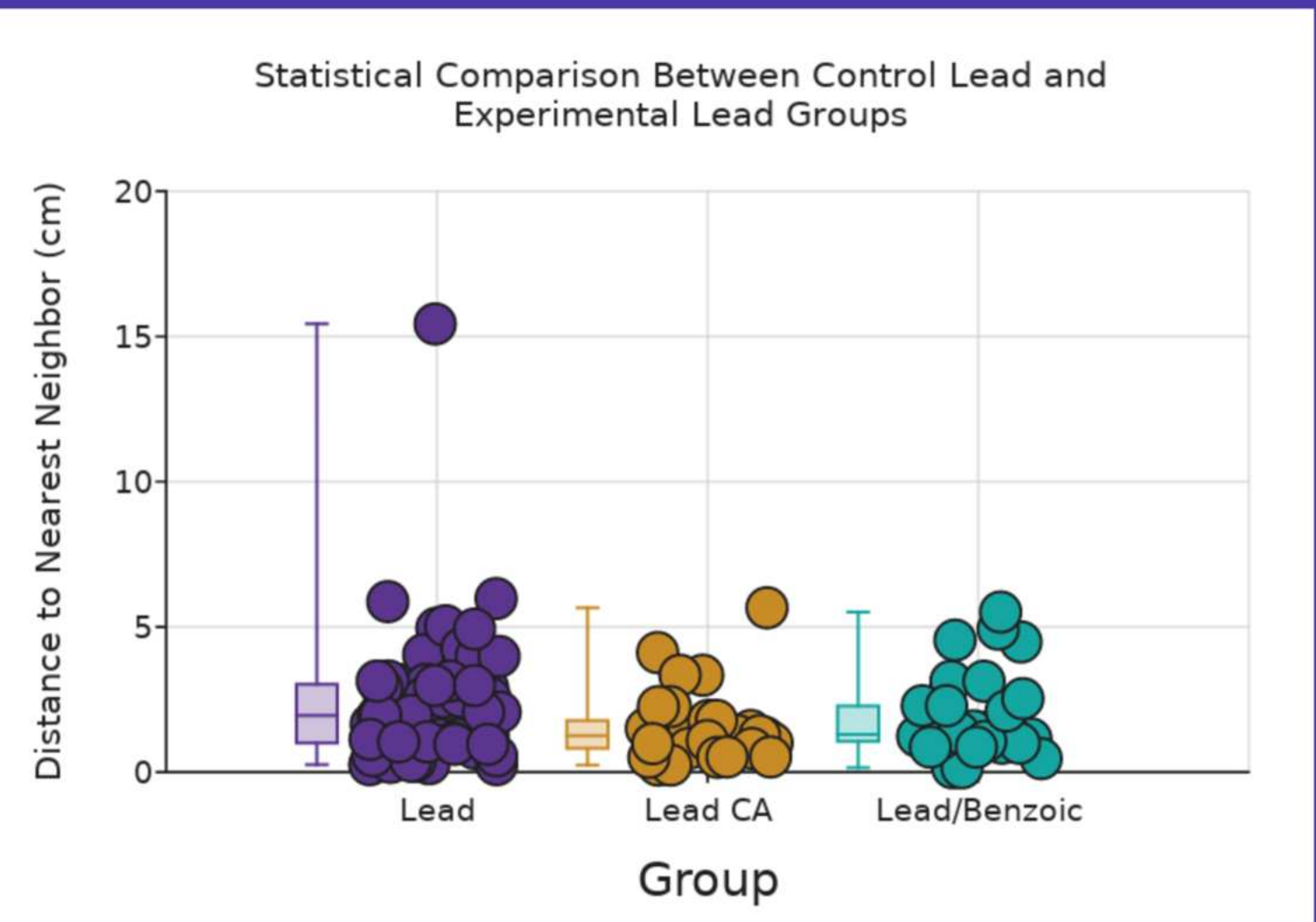
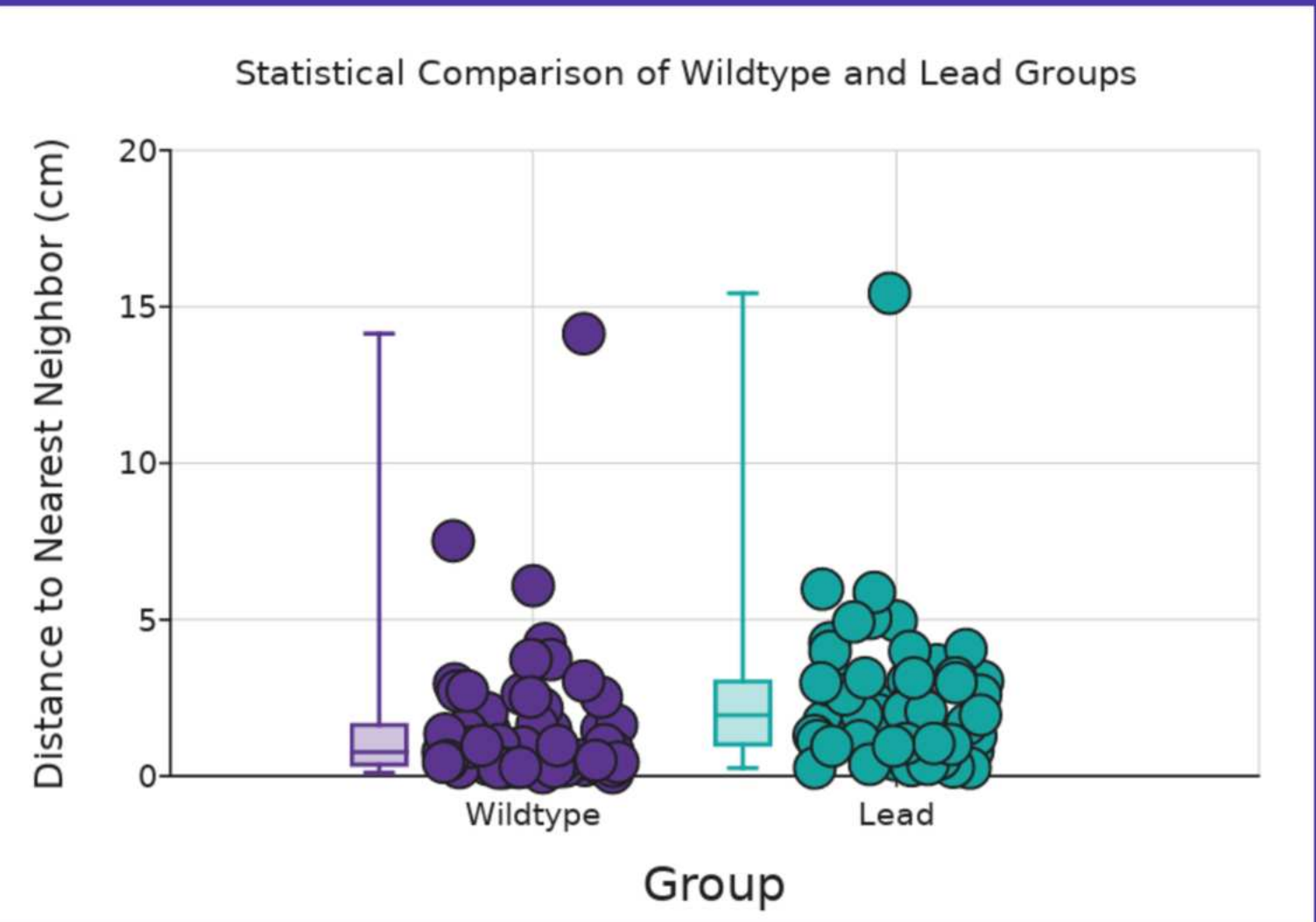


## Data



The raw data from all control and experimental groups. The data indicates that the control wildtype flies tend to stay within 0.5cm of their nearest neighbor while the flies from the experimental groups generally tend to stay more than 1 cm away from their nearest neighbor.

# Results & Analysis



Social interaction is a **common symptom** of ASD and can be observed by **measuring the distance** between each fly and their nearest neighbor. The greater the distance between neighboring flies, the lower the social interaction. The results along with the raw data indicate the the **method to induce** the *Drosophila* with ASD-like symptoms had the **desired effect**. This is seen in the collected data since the **lead group had greater distances** between flies compared to the wildtype group.

Regarding the **citric acid diet**, the results indicate that the citric acid treatment on the ASD-induced flies **did not have the expected effect**. This unexpected results is likely due to citric acid's **antioxidative properties** which would have aided in **alleviating** the lead-induced neurotoxicity. For the **benzoic acid diet**, it was concluded that the supplement had **no effect** on the level of social impairment exhibited by the ASD-induced flies. This is likely due to benzoic acid **not having antioxidant properties** to combat neurotoxic events.

## Limitations

- Small sample size: only ten flies tested per trial
- Lack of trials: only three trials were conducted per group
- Reproduction rate of experimental flies

## Implications

The results from this research aim to aid individuals affected by ASD in planning out their diets. The results also serve to help these individuals to easily identify foods that could potentially alleviate or aggravate the symptoms they experience.

## References

Centers for Disease Control and Prevention. (2022, December 9). What is autism spectrum disorder? <https://www.cdc.gov/ncbddd/autism/facts.html>

Hagerman, R. J. (2020). Fragile X is a common cause of autism and intellectual disability. *News*. <https://health.ucdavis.edu/news/health-wellness/fragile-x-is-a-common-cause-of-autism-and-intellectual-disability/2020/10/>

Lamas, A., Regalado, V., Vazquez, S., Cordero, A., Franco, C.M. Short Chain Fatty Acids Commonly Produced by Gut Microbiota Influence Salmonella enterica Motility, Biofilm Formation, and Gene Expression. *Antibiotics* (Basel). 2019 Dec 13;8(4):265. doi: 10.3390/antib8040265. PMID: 31847278; PMCID: PMC6963744.

MacFabe, D.F. (2012) Short-chain fatty acid fermentation products of the gut microbiome: implications in autism spectrum disorders. *Microbial Ecology in Health and Disease*, 23(1). DOI: 10.3402/mehd.v23i1.19290

Mansury, A., Aubert, V. and Ravussin, V. (2023) Short chain fatty acids: the messengers from down below. *Front. Neurosci.* 17:1197759. doi: 10.3389/fnins.2023.1197759

McNeil, A.S., Jolley, S.N., Akhryeva, A.A., Hurlbut, M., Rozycki, T., Milunovich, A.J., Chambers, M.C., Simon, A.F. Conditions Affecting Social Space in *Drosophila melanogaster*. *J Vis Exp*. 2016 Nov 5;(105):e53242. doi: 10.3791/53242. PMID: 26920306; PMCID: PMC4929198

National Institute of Mental Health. (2024, February). Autism spectrum disorder. <https://www.nimh.nih.gov/health/topics/autism-spectrum-disorders-asd>

Seigneur, L., Guzik, Y., Blad, R., Shaw, P.J. Aversive phototoxic suppression: evaluation of a short-term memory assay in *Drosophila melanogaster*. *Genes Brain Behav*. 2009 Jun;8(4):377-88. doi: 10.1111/j.1469-7580.2009.02045.x. Epub 2009 Feb 11. PMID: 19220476; PMCID: PMC4094202

Simon, A.F., Chou, M.T., Solari, S.D., Nicholas, T., Saini, N., Melchior, S., Krantz, D.E. A simple assay to study social behavior in *Drosophila*: measurement of social space within a group. *Genes Brain Behav*. 2012 Mar;11(2):243-52. doi: 10.1111/j.1469-7580.2011.02470.x. Epub 2011 Nov 23. PMID: 22010812; PMCID: PMC3268943

Shih, C., Anagnostis, E.P., Anthony, A., & Goulet, A. (2021). Lead (Pb) induced oxidative stress as a mechanism to cause neurotoxicity in *Drosophila melanogaster*. *Toxicology*, 462, 152999. <https://doi.org/10.1016/j.tox.2021.152999>

Zauderer, S. (2023, September 18). Autistic Brain vs Normal Brain. *Crossrivertherapy.com*. Cross River Therapy. <https://www.crossrivertherapy.com/autism/autistic-brain-vs-normal-brain/differences-in-brain-connectivity>